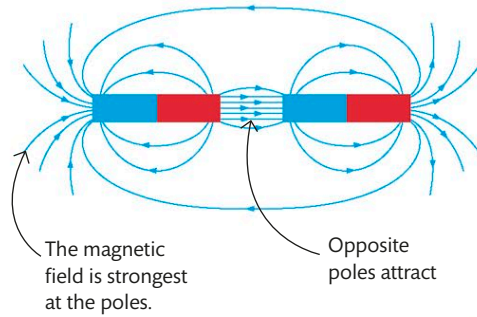
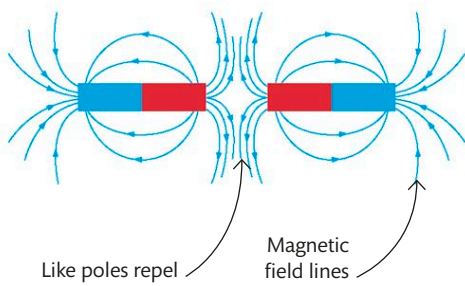


MAGNETS

Magnets are objects that attract other magnets or things containing iron or other magnetic metals, such as nickel and cobalt. A magnet has two ends called a north and south pole. When opposite poles of a magnet are held near each other, they attract. When the same poles are held close to each other, they repel. The area around a magnet, in which magnetic forces are exerted, is called a magnetic field.



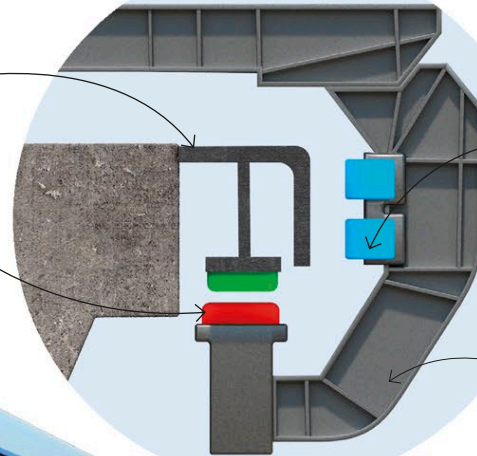
ELECTROMAGNETS

An electromagnet is made when electricity is passed through a wire, often coiled around an iron core, causing it to produce a magnetic field. Huge electromagnets are used to lift heavy objects, such as scrap metal.

CLOSE-UP OF TRAIN MECHANISM

The guiding rail contains wire coils that produce a changing magnetic field.

Magnets on the underside of the train repel the magnets in the guiding rail, causing the train to levitate.



The magnets on the side of the train are in turn repelled and attracted by the magnetic field of the guiding rail, causing the train to move.

The side of the train wraps around the guiding rail.

